

Rでの解析の流れ

```
> 18-11A_syusyoku <- read.table('./syusyoku_jyuuyou.csv', header=T, sep=',')
> head(18-11A_syusyoku)
  ID yuumei shinjin wakate senmon dentou ryuugaku.shingaku nobisou
1  A      2      5      1      5      1              5      2
2  B      3      4      2      5      3              4      1
3  C      4      1      1      2      5              3      2
4  D      4      1      3      2      5              3      4
5  E      1      2      5      1      2              1      4
6  F      5      1      1      1      4              2      2

> 18-11A_syusyoku_fact <- factanal(18-11A_syusyoku[,2:8], factors=3, scores='regression', rotation='varimax')
> 18-11A_syusyoku_fact

Call:
factanal(x = df[, data_cols], factors = factors, scores = scores, rotation = rotation)

Uniquenesses:
              yuumei          shinjin          wakate          senmon
              0.300          0.125          0.005          0.005
dentou ryuugaku.shingaku nobisou
0.094          0.199          0.346

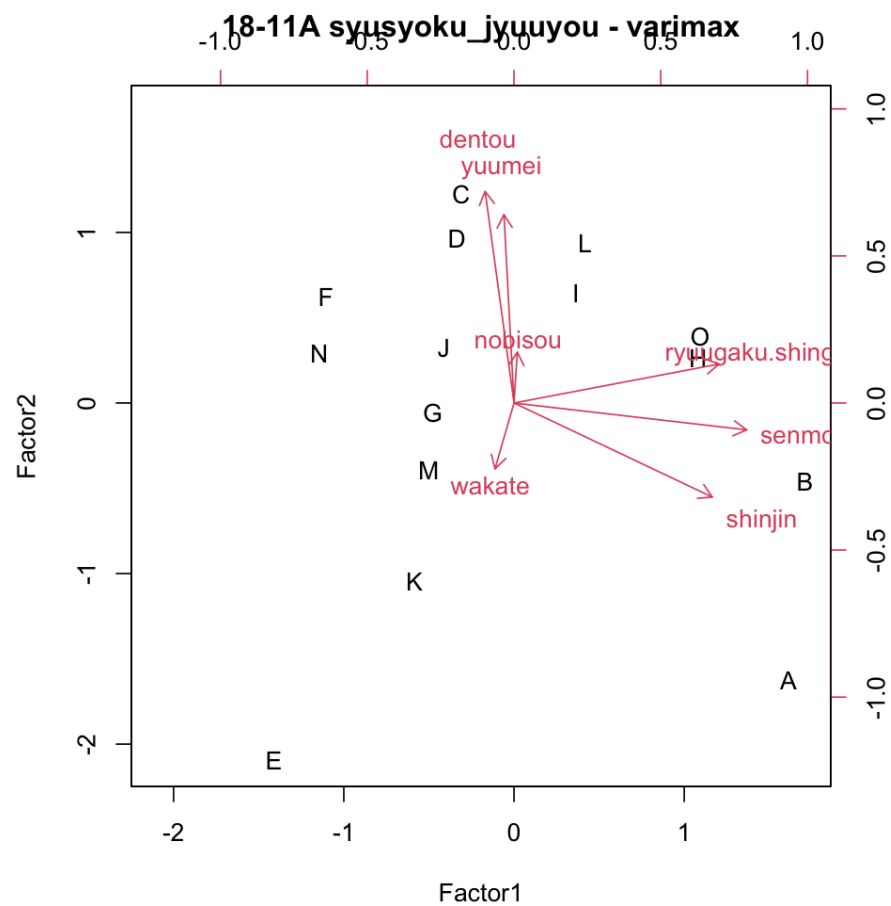
Loadings:
              Factor1 Factor2 Factor3
yuumei              0.800 -0.241
shinjin             0.845 -0.400
wakate              -0.280  0.954
senmon              0.991 -0.114
dentou             -0.123  0.899  0.287
ryuugaku.shingaku  0.873  0.166 -0.105
nobisou              0.216  0.780

              Factor1 Factor2 Factor3
SS loadings    2.482    1.774    1.671
Proportion Var  0.355    0.253    0.239
Cumulative Var  0.355    0.608    0.847

Test of the hypothesis that 3 factors are sufficient.
The chi square statistic is 8.22 on 3 degrees of freedom.
The p-value is 0.0417

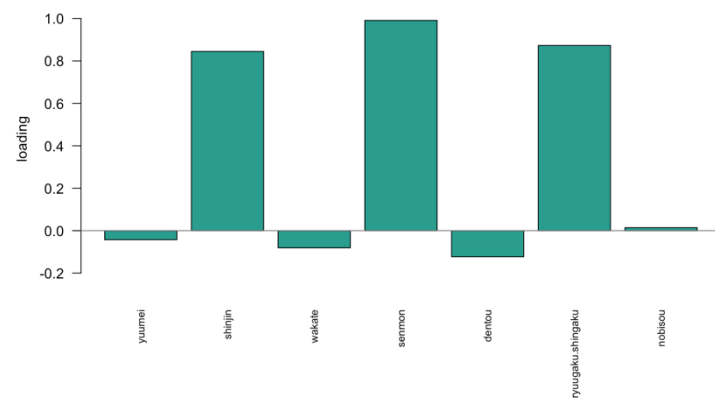
> 18-11A_syusyoku_fact$loadings[,1:3]
              Factor1    Factor2    Factor3
```

因子得点と因子負荷量



因子負荷量の棒グラフ

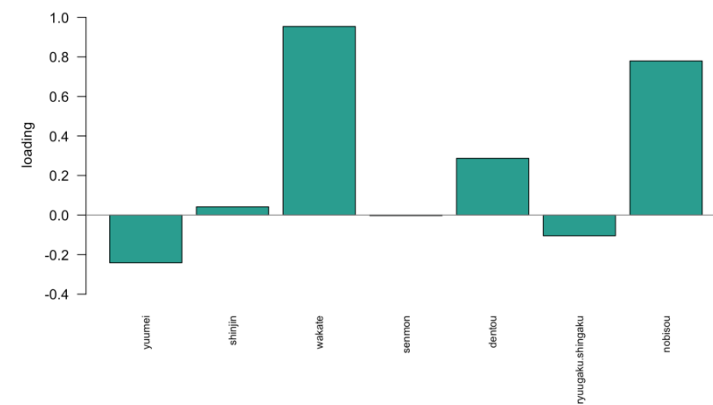
18-11A syusyoku_jyuuyou - varimax Factor1 loadings



18-11A syusyoku_jyuuyou - varimax Factor2 loadings



18-11A syusyoku_jyuuyou - varimax Factor3 loadings



ソースコード (1/2)

Class13_rp18-11A_02.R

```
1 setwd("/Users/kakigoori/Desktop/data/13")
2
3 dir.create("./answers/plots_pdf", recursive=TRUE, showWarnings=FALSE)
4
5 syusyoku <- read.table("./syusyoku_jyuuyou.csv", header=T, sep=",")
6
7 syusyoku
8
9 head(syusyoku)
10
11 syusyoku_fact <- factanal(syusyoku[,-1], factors=3, scores="regression", rotation="varimax")
12
13 str(syusyoku_fact)
14
15 syusyoku_fact
16
17 syusyoku_fact$loadings[,1:3]
18
19 apply(syusyoku_fact$loadings, 1, function(x){1-sum(x^2)})
20
21 syusyoku_fact$scores
22
23 pdf("./answers/plots_pdf/18-11A_syusyoku_biplot.pdf", width=8.5, height=6.2)
24 biplot(syusyoku_fact$scores, syusyoku_fact$loadings, xlab=syusyoku[,1])
```

ソースコード (2/2)

Class13_rp18-11A_02.R

```
25 dev.off()
26
27 pdf("./answers/plots_pdf/18-11A_syusyoku_factor1_bar.pdf", width=8, height=5.2)
28 barplot(syusyoku_fact$loadings[,1], cex.names=0.8, las=2, ylim=c(-0.5,1.1))
29 abline(h=0, col="gray50")
30 dev.off()
31
32 pdf("./answers/plots_pdf/18-11A_syusyoku_factor2_bar.pdf", width=8, height=5.2)
33 barplot(syusyoku_fact$loadings[,2], cex.names=0.8, las=2, ylim=c(-0.5,1.1))
34 abline(h=0, col="gray50")
35 dev.off()
36
37 pdf("./answers/plots_pdf/18-11A_syusyoku_factor3_bar.pdf", width=8, height=5.2)
38 barplot(syusyoku_fact$loadings[,3], cex.names=0.8, las=2, ylim=c(-0.5,1.1))
39 abline(h=0, col="gray50")
40 dev.off()
```